# TUTORIAL 4

Design For Test

## SUBMITTED BY

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# 1 Goals

The goal of this lab is to learn the basic ideas of Design for Test (DFT) and how to create automatic test patterns for digital circuits. To keep things simple, we work with a small 8-bit adder. This allows us to focus on the DFT flow without dealing with a complicated design.

In this experiment, we performed four main tasks:

1. Modify the design
2. Integrate scan architecture into the design
3. Generate test patterns using ATPG
4. Simulate the design with the generated patterns

# 2 Design the unit (RTL)

In this part of the lab, we created the RTL file, 8-bit registered adder, `my_adder.vhd` in the directory `dft/rtl`.

After the basic design was done, we modified the entity to prepare it for scan insertion. We added the three scan-related ports:

- SERIAL_IN
- SCAN_EN
- SERIAL_OUT

# 3 Lunch Design Compiler

Before starting the synthesis tool, we created a `.synopsys_dc.setup` file to select the standard cell library and defined it in log file directories. This helps to organize all generated data properly.

then, we defined the library path and chose `NangateOpenCellLibrary_typical_ccs.db` as the target library, which contains typical delay values.

we ran the command `design_vision | tee log/synthesis.log &` for launch the tool. Here, `design_vision` command executes to launch the tool and tee log helps to save the log file to the log directory and it would save by name of synthesis.log.

# 4 DFT Configuration and Synthesis

## 4.1 Design Import

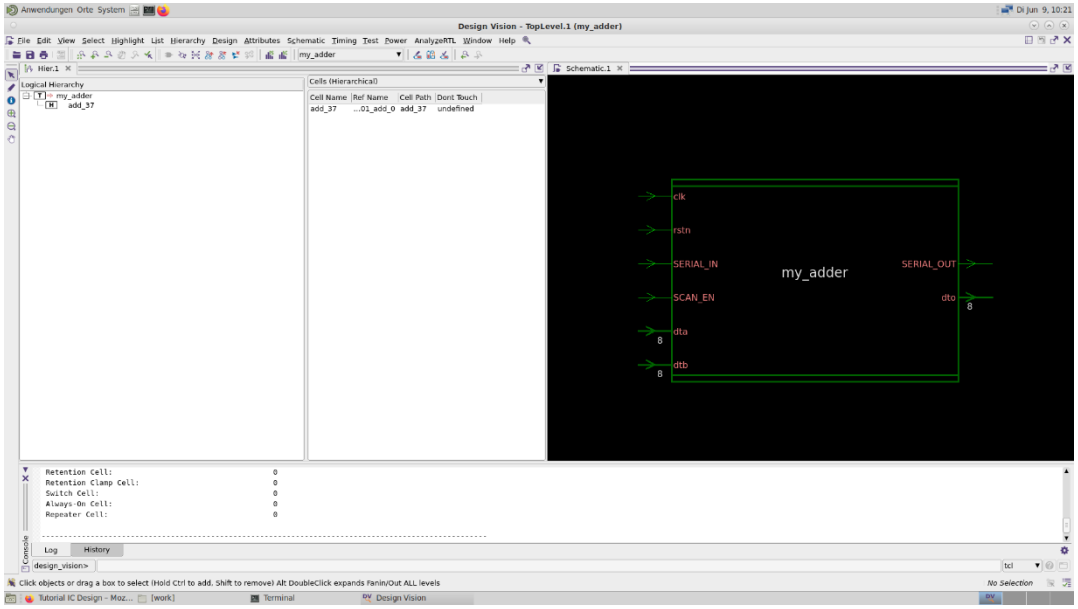
In the Design Vision prompt, we ran the command `read_file -format vhdl {../rtl/my_adder.vhd}` to read the RTL file that we wanted to synthesize. After executing

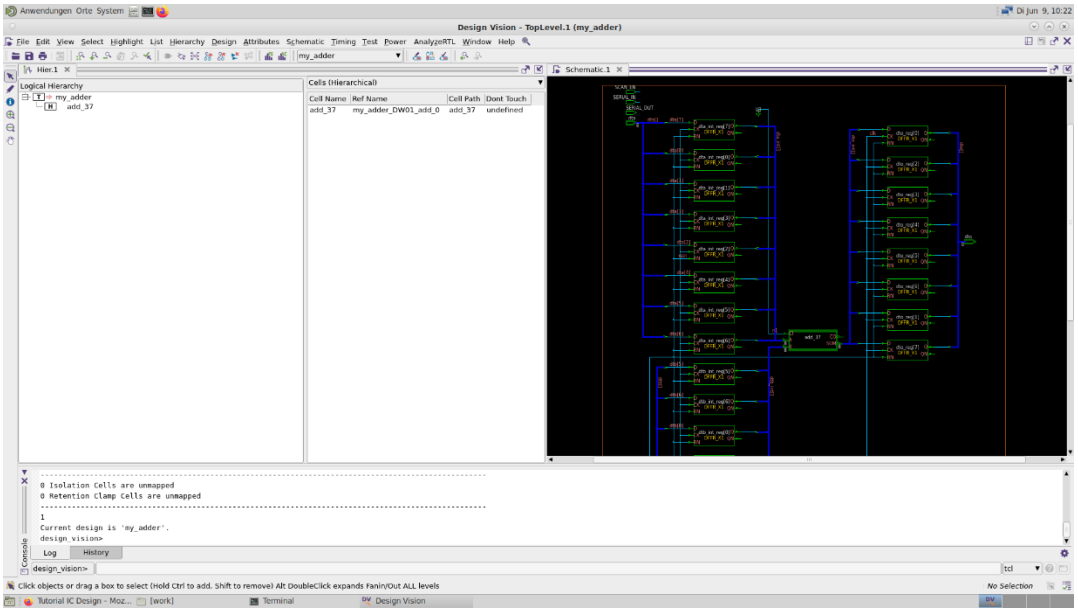
this command successfully, the tool showed some reports. Unfortunately, I forgot to take the report screenshot. Though it looks similar to the report that given in the manual. So, I took it from the manual.

```
Inferred memory devices in process
        in routine my_adder line 22 in file
                '/usrf06/home/agids/usr/agarcia/GIT/tutorial_ic_design/my_adder/rtl/my_adder.vhd'.
===============================================================================
|    Register Name    |   Type    | Width | Bus | MB | AR | AS | SR | SS | ST |
===============================================================================
|       dto_reg       | Flip-flop |   8   |  Y  | N  | Y  | N  | N  | N  | N  |
|     dta_int_reg     | Flip-flop |   8   |  Y  | N  | Y  | N  | N  | N  | N  |
|     dtb_int_reg     | Flip-flop |   8   |  Y  | N  | Y  | N  | N  | N  | N  |
===============================================================================
Presto compilation completed successfully.
```

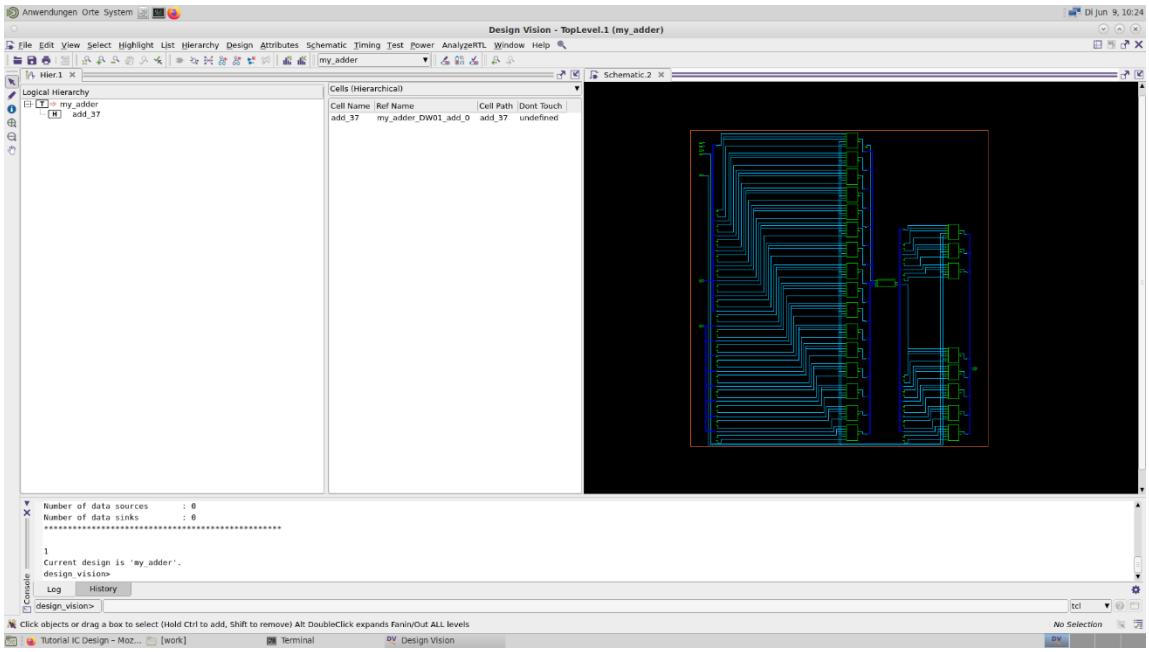
## 4.2 DFT synthesis

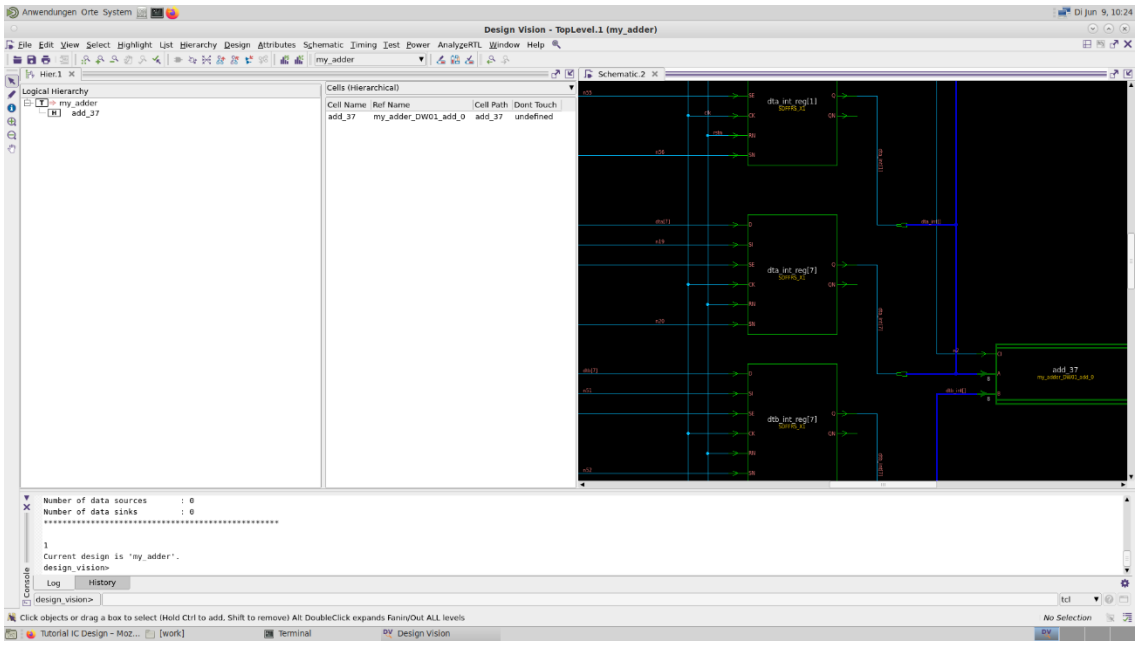
First, we compiled the design in Design Compiler and checked the schematic. We observed that the tool used `DFFR_x1` flip-flops in the design.



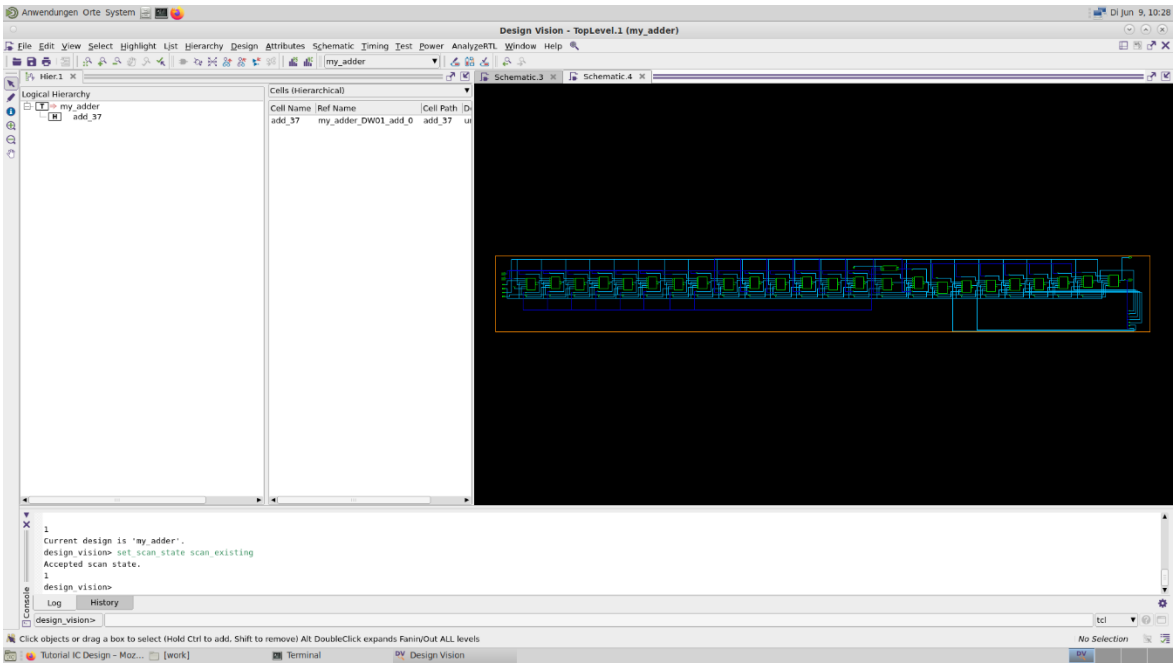


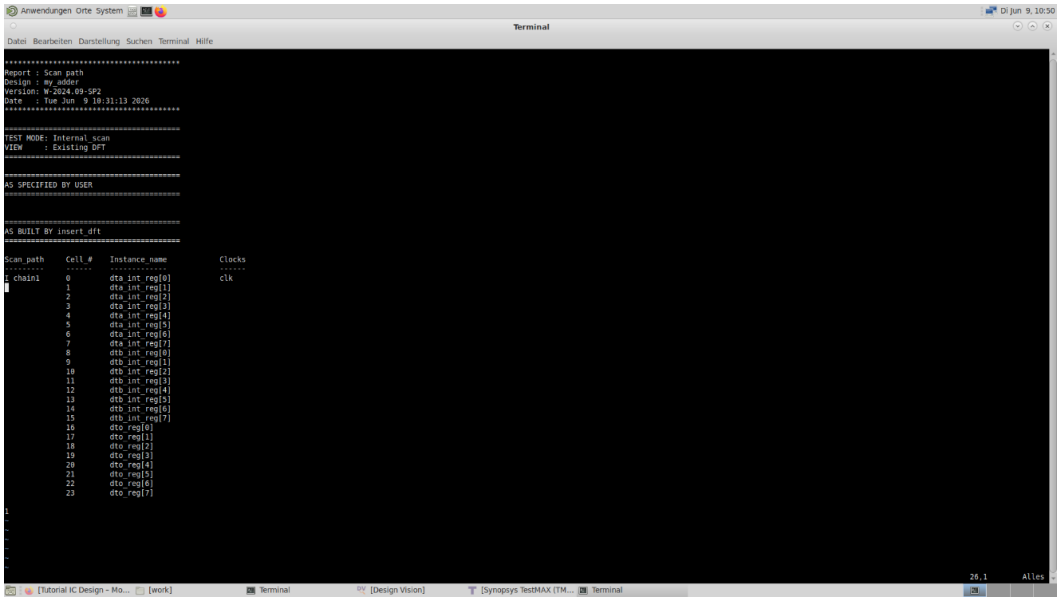

Then We set up the DFT configuration: scan style, test period, clock, reset, scanDatain, scanDataout, and scan-enable. After that, we ran `compile -scan`, and the tool replaced the flip-flops with `SDFFRS_X1` scan cells.

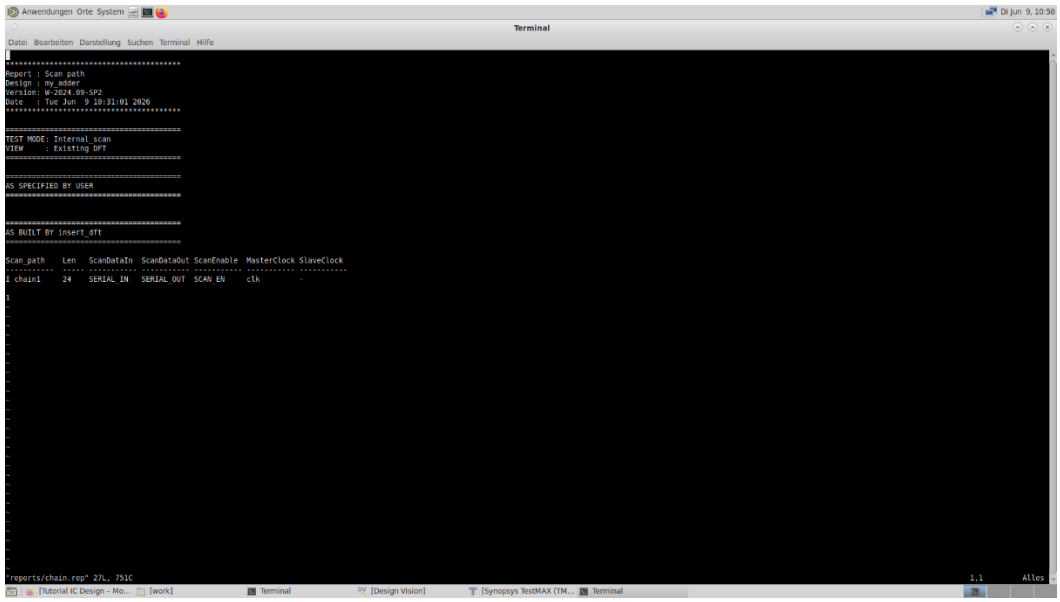





We previewed the scan structure, checked DFT rules, defined one scan chain, set the scan path, and inserted the scan chain. To confirm that the scan chain is implemented or not, we ran the command `set_scan_state scan_existing`. We confirmed again that the scan chain is added and we produced some reports including the scan chain and the instances in each chain using the `report_scan_path`.






Next, we generated reports using commands like `report_area` and `report_power`, and saved them in the report's directory.

```
****************************************

Information: Updating design information... (UID-85)
Library(s) Used:

    NangateOpenCellLibrary (File: /usrf07/home/agids/lab/l_ids309/0_FreePDK45/LIB/NangateOpenCellLibrary_typical_cc

Number of ports:                          55
Number of nets:                          108
Number of cells:                          59
Number of combinational cells:             9
Number of sequential cells:               24
Number of macros/black boxes:              0
Number of buf/inv:                         0
Number of references:                      2

Combinational area:                32.452000
Buf/Inv area:                       0.000000
Noncombinational area:            185.136005
Macro/Black Box area:               0.000000
Net Interconnect area:      undefined  (Wire load has zero net area)

Total cell area:                  217.588006
Total area:                 undefined
1
design_vision>
```

```
i - Including register clock pin internal power


  Cell Internal Power  =  22.8538 uW   (95%)
  Net Switching Power  =   1.3208 uW    (5%)
                         ---------
Total Dynamic Power    =  24.1746 uW  (100%)

Cell Leakage Power     =   3.1086 uW

Information: report_power power group summary does not include estimated clock tree power. (PWR-789)

                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000            0.0000           0.0000  (   0.00%)
memory             0.0000           0.0000            0.0000           0.0000  (   0.00%)
black_box          0.0000           0.0000            0.0000           0.0000  (   0.00%)
clock_network      0.0000           0.0000            0.0000           0.0000  (   0.00%)  i
register           0.0000           0.0000            0.0000           0.0000  (   0.00%)
sequential        21.9436           0.9921        2.6962e+03          25.6318  (  93.95%)
combinational      0.9103           0.3287          412.4165           1.6514  (   6.05%)
--------------------------------------------------------------------------------------------------
Total             22.8538 uW         1.3208 uW     3.1086e+03 nW        27.2832 uW
1
design_vision>
```

Finally, we saved the netlist using `write_file` command in Verilog formats after applying `change_names` rules. We also saved the scan_def file and test protocol file.

# 5 ATPG

## 5.1 Lunch TetraMAX

In this part of the lab, We used Synopsys TetraMAX to generate test patterns for the scan-inserted adder design. We used `tmax` to launch the tool.

## 5.2 Read the netlist and library files

In this step, we loaded the files needed for ATPG in TetraMAX. We added the gate-level netlist file that we got from the synthesis output as well as the library definition of each cell in Verilog format.

## 5.3 Build the design

After loading the netlist and the library, We opened the Build Model dialog and selected my top module name, `my_adder`. We used the default learning options. TetraMAX then created a new in-memory simulation model, deleting old model automatically.

## 5.4 DRC Check

In this step, we performed the Design Rule Check in TetraMAX. We selected the STIL procedures file, which contains the scan-chain and test-mode definitions. After choosing the file, we clicked Run to execute the DRC. TetraMAX checked the design for any test-related rule violations before ATPG.

## 5.5 Generate the test pattern

After finishing the DRC with no errors, we generated the test patterns in TetraMAX. We added all stuck-at faults to the design and then ran automatic ATPG.

TetraMAX created 7 test patterns and achieved 100% stuck-at fault coverage. The report showed all detectable faults were covered, with no remaining undetected faults.

```
TEST-T> add_faults -all
  378 faults were added to fault list.
TEST-T> run_atpg -auto

  ATPG performed for stuck fault model using internal pattern source.
  ---------------------------------------------------------
  #patterns     #faults     #ATPG faults  test      process
  stored     detect/active  red/au/abort  coverage  CPU time
  ---------  -------------  ------------  --------  --------
  Begin deterministic ATPG: #uncollapsed_faults=154, abort_limit=10...
  7             154      0         0/0/0   100.00%      0.01

      Uncollapsed Stuck Fault Summary Report
  -----------------------------------------------
  fault class                    code   #faults
  -----------------------------  ----  ---------
  Detected                        DT        354
  Possibly detected               PT          0
  Undetectable                    UD         24
  ATPG untestable                 AU          0
  Not detected                    ND          0
  -----------------------------------------------
  total faults                              378
  test coverage                          100.00%
  -----------------------------------------------
            Pattern Summary Report
  -----------------------------------------------
  #internal patterns                          7
      #basic_scan patterns                    7
  -----------------------------------------------
            CPU Usage Summary Report
  -----------------------------------------------
  Total CPU time                           0.17
  -----------------------------------------------
```